A Borrowed Recovery?

The 2025 Teacher-Workforce Data for the United States

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American teachers have stopped leaving, and American teaching has not recovered. Both things are true in the same year, in the same data, and the distance between them is the story of 2025.

The first is easy to verify. In the year to 2025, 7.9 percent of public school teachers left education, almost exactly the 7.7 percent that was normal in the fifteen years before the pandemic; the spike of the COVID years, 10.4 percent at its worst, was already gone by 2021 (Figure 2). Across two decades of linked survey data, ninety thousand teacher-years of them, the exodus turns out to have been a two-year fever rather than a new era, and the fever broke four years ago.

The second is the subject of this essay. Nothing that made teachers leave has been repaired. Pay stands further behind the graduate labor market than at any point since 2010, teachers are abandoning the classroom for the jobs around it at the fastest rate on record, the typical new recruit is a forty-two-year-old career changer rather than a young graduate, and in 2025 the workforce shrank for the first time since the pandemic trough. What holds retention in place is none of the things that usually hold a profession together; it is the coldest hiring market in a decade, quietly doing the job that pay and working conditions were supposed to do. The calm is rented, and the lender is the business cycle.

Fewer pupils, and now fewer teachers

American public schools have been losing students since the fall of 2019, when enrollment peaked at 50.8 million; smaller birth cohorts and the pandemic exodus have since removed 1.4 million pupils, and the official projections point further down. For five years the teaching workforce paid no attention, dipping in 2021, recovering, and standing by 2024 nearly where it had stood at the peak. The result was a quiet improvement in pupil-teacher ratios that nobody designed (Figure 1).

In 2025 the workforce finally caught up with its students, falling from 4.54 to 4.37 million qualified teachers, a loss of roughly 170,000 and the sharpest one-year decline in the series outside the pandemic itself. Some of this had to come, because districts staffed up on federal relief money that has now run out, and letting attrition shrink the payroll is how a school system downsizes without ever pronouncing the

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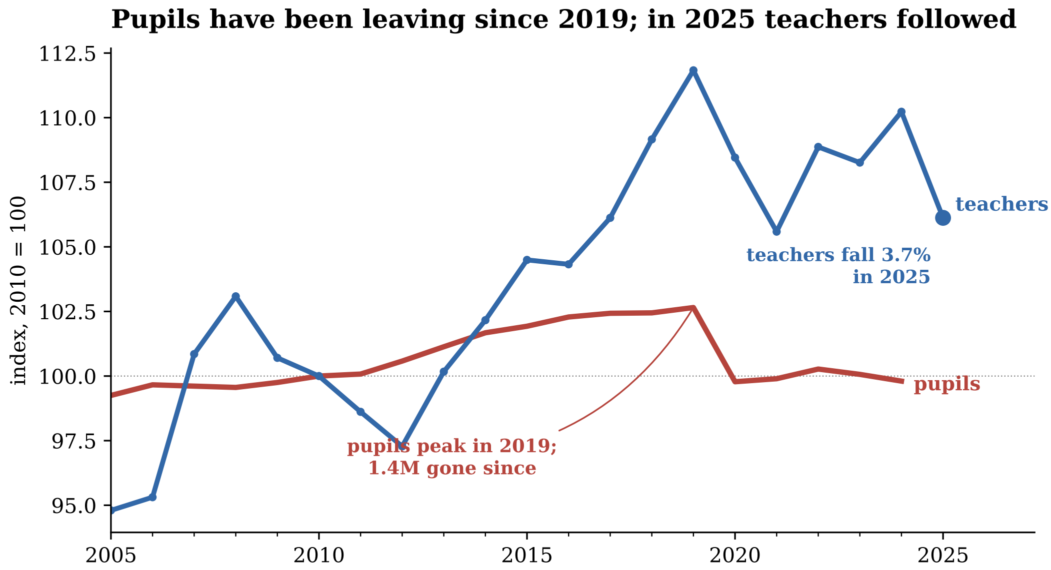

Figure 1: Public-school pupils and full-time qualified teachers, indexed to 2010.

Teachers: CPS monthly data, employed K–12 teachers (occupations 2310–2330) with at least a bachelor's degree, annual average of weighted monthly stocks; survey estimates carry sampling error of roughly ±1–2 percent. The 2025 average covers eleven months, because no CPS was collected in October 2025 during the federal shutdown; the like-for-like eleven-month figure for 2024 (4.52M) confirms the decline is not an artefact. Pupils: NCES, Digest of Education Statistics, table 203.10, fall enrollment in public schools.

word layoff. A workforce falling in step with its student body is not, on its own, a problem. Whether it becomes one depends on who is leaving and who is arriving, which is what the rest of the data describe.

Two ways to leave a classroom

There are two honest ways to count a lost teacher. The narrow way asks whether she still works anywhere in education; the broad way asks whether she is still teaching a class. The distinction sounds like bookkeeping, and it carries the whole story of the year (Figure 2).

Counted the narrow way, the drama was brief. Attrition from education spiked to 10.4 percent in the year the pandemic hit, was nearly back to its average by 2021, and has stayed there since; the age profile has settled with it, the youngest teachers included (Figure 3). This is the line the celebrations are reading, and read correctly it says the exodus ended years ago.

Counted the broad way, nothing has healed. Last year 13.3 percent of public school teachers left the classroom, roughly two points above the old norm, and the wedge between leaving the classroom and leaving education reached 5.3 percentage points, the widest in the twenty years I can measure, against a pre-pandemic average of 3.4. Two of every five teachers who left a classroom in 2024 stayed in education, as instructional coaches, deans, counselors, tutors or college instructors.

Note that this is exactly the pattern a cold outside market should suppress, unless the competition for teachers now sits inside the sector itself. It does: the relief money that financed the hiring boom also financed an expansion of coaching, intervention and administrative roles, jobs that offer the identity of education without the classroom, and moves into them are not disciplined by the private labor market at all. For the sector's payroll such moves are churn, invisible in the statistics that make headlines. For the

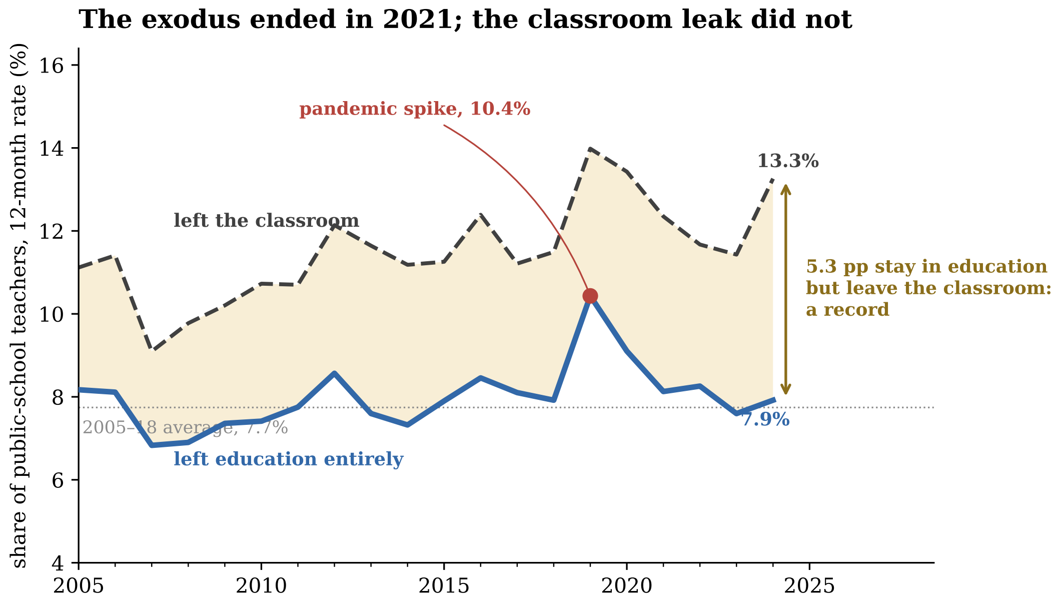

Figure 2: Two ways to leave: out of the classroom, and out of education.

Public-school full-time K–12 teachers with a bachelor's degree or more, followed twelve months in linked CPS interviews, 2005–2024 base years ($n = 91{,}035$ teacher-years; 2,841 in 2024, which lacks October links because no October 2025 survey exists). Blue: working in no education occupation at $t{+}12$ and not teaching in any later interview. Dashed: no longer a classroom teacher by the same standard. The marked 2019 point links into 2020 and carries the first COVID wave. The gap between the lines is movement into other education work.

school it is the same empty desk at the front of the same room, except that the person who used to stand at it is now down the hall, earning slightly more to teach nobody.

The new teacher is forty-two

Attrition is only half of the ledger; the other half is replacement, and it is the half that looks least like a recovery. Entries did not keep pace with exits in 2024–25, which is the arithmetic behind the shrinking stock of Figure 1, and the profile of those who did arrive is telling. The median teacher who entered an American classroom between 2024 and 2025 was forty-two years old; three quarters came straight from another job, and fewer than one in five was under thirty. The profession is aging from the entry door, with teachers under thirty-five down from 32 percent of the workforce in the mid-2000s to 29 percent today.

None of this means career changers make worse teachers, and a profession that a forty-two-year-old engineer still wants to join is healthier than one nobody re-enters. Note also that about one apparent leaver in six is back in a classroom within months, a revolving door that quietly softens every flow in this essay.

But consider what the composition implies. Hire a twenty-five-year-old and you buy, in expectation, three decades of future teaching; hire a forty-two-year-old and you buy half of that. And a career changer is, by construction, someone comparing two careers in real time, which makes the inflow exquisitely sensitive to the one variable that has moved steadily in the wrong direction.

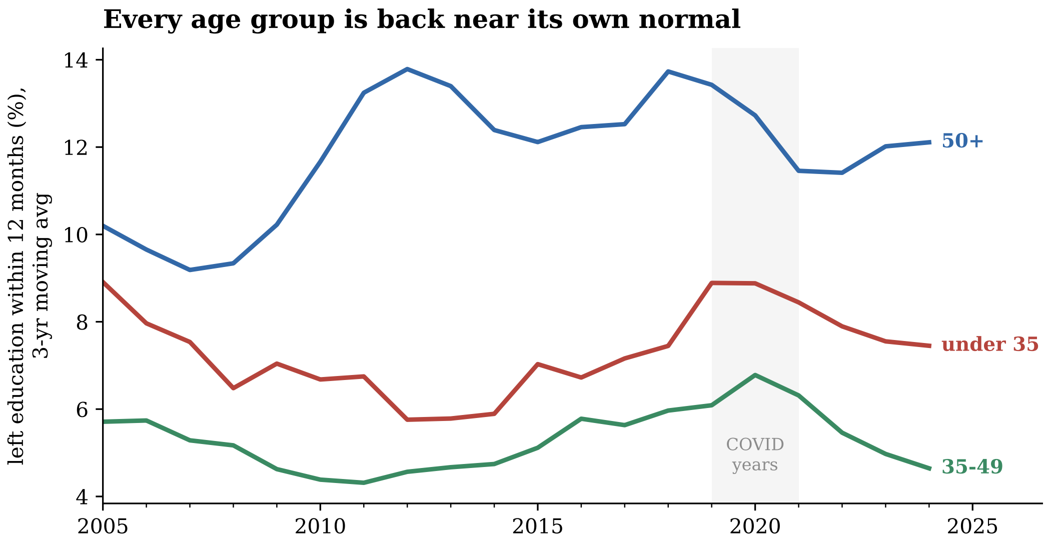

Figure 3: Attrition from education by age of teacher.

Same universe and definition as the blue line of Figure 2, split by age at the base interview; three-year centered moving averages. The 50+ line embeds retirements and is structurally the highest.

The pay that never came back

A teacher at the median earned $1,380 a week in 2025, in dollars of that year; her counterpart in 2010 earned $1,462. Fifteen years of budgets, strikes and salary schedules have therefore left the median American teacher nearly 6 percent poorer in real terms, over a period in which the median full-time worker gained 9 percent and the median college graduate gained 3 (Figure 4). Measured against the graduate market in which teaching actually recruits, teacher pay has slid from 87 cents on the graduate dollar to 79. The inflation of 2021 to 2023, which salary schedules were too rigid to match, did most of the damage, and the rebound since has recovered less than half of what those two years took.

So the puzzle is this. Teaching pays relatively worse than at any point since 2010, the classroom leaks into the back office at a record rate, and attrition from the sector nevertheless sits on its long-run average. Something is holding teachers in place, and the pay data say plainly that it is not the pay.

A recovery on loan

Teachers are people, and people quit jobs when there are other jobs to go to. Figure 5 plots the rate at which teachers leave education against the national unemployment rate, drawn on an inverted scale so that good times point upward, and the two lines breathe together. When unemployment reached ten percent after 2008, teacher attrition sat at the floor of its range for years; as the long expansion of the 2010s tightened, it climbed; it peaked at its all-time high in the 2019 base year, when the labor market was the tightest in half a century, though that observation links into 2020 and so carries the first COVID wave as well. It has now come down alongside a cooling market.

The same relationship shows up sharper against the private-sector quits rate, the economy's willingness to walk out of a job: about 0.6 percentage points of teacher exit per point of quits, a loose but persistent link across two decades (Figure 6). And quits have collapsed, from 3.1 percent a month at the height

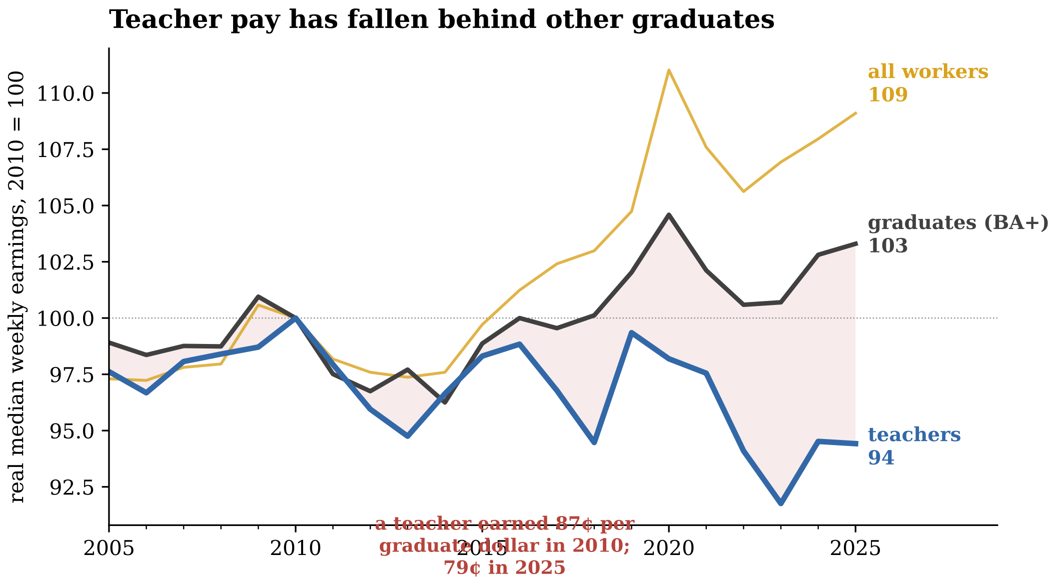

Figure 4: Real median weekly earnings: teachers, all graduates, all workers.

Teachers: weighted median usual weekly earnings of full-time K–12 teachers (BA+), CPS outgoing-rotation interviews, deflated by CPI-U. Graduates: BLS median usual weekly earnings, bachelor's degree and higher, age 25+. All workers: BLS real median usual weekly earnings, age 16+. Indexed to 2010 = 100.

of the 2022 frenzy to 2.3, with hiring at its weakest in a decade and vacancies per unemployed worker below their pre-pandemic level. Run that 2.3 through the historical relationship and it predicts teacher attrition a shade above 8 percent; the observed rate is 7.9. Nothing in the retention of 2025 requires any improvement in teaching to explain it, which is the polite way of saying there has been none.

That is the loan. The calm in teacher retention is collateralized by the weakness of everyone else's job market, and the same fitted line implies that a return of quits to their 2022 level would push attrition back up by around half a point, with the young and the career changers, the margins already thinnest, moving first. What the labor market has lent, the labor market can recall.

The window

Read quickly, the 2025 data close the book on the pandemic teacher crisis, with attrition normal, every age group settled and ratios improving. Read carefully, they describe a workforce in remission rather than in health: a stock that is now falling, a classroom that leaks into the jobs beside it at a record rate, new supply that is old and thin among the young, and pay that is eight points further behind the graduate market than in 2010, the whole arrangement stabilized by nothing more durable than a slow economy.

School systems control exactly one of those variables. A slack labor market is the cheapest possible moment to restore the relative pay of teachers, and to make the classroom a job worth keeping rather than a stepping stone to the jobs around it, precisely because retention does not need rescuing at the same time. The next boom will re-run the experiment of 2019 on an older, thinner, worse-paid workforce, unless someone uses the interval to change the terms. The 2025 data are not the end of the story. They are the intermission.

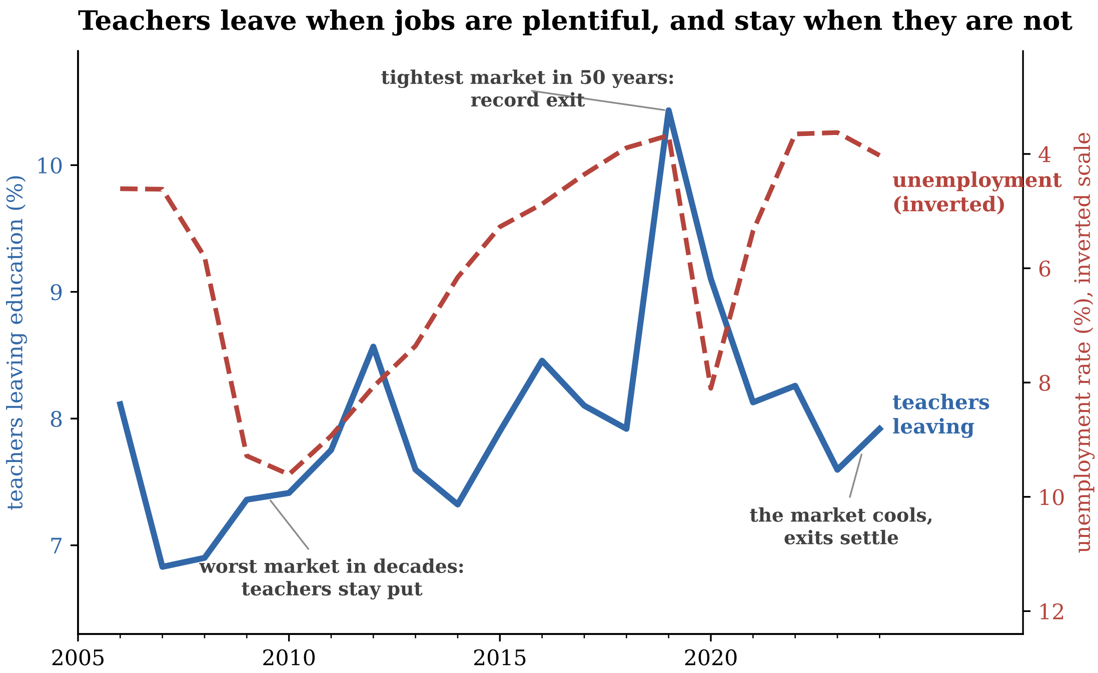

Figure 5: Unemployment and the rate at which teachers leave education.

Teachers leaving education as in Figure 2; national unemployment rate (annual average of monthly rates, BLS), plotted on an inverted right axis so that a hotter labor market plots higher.

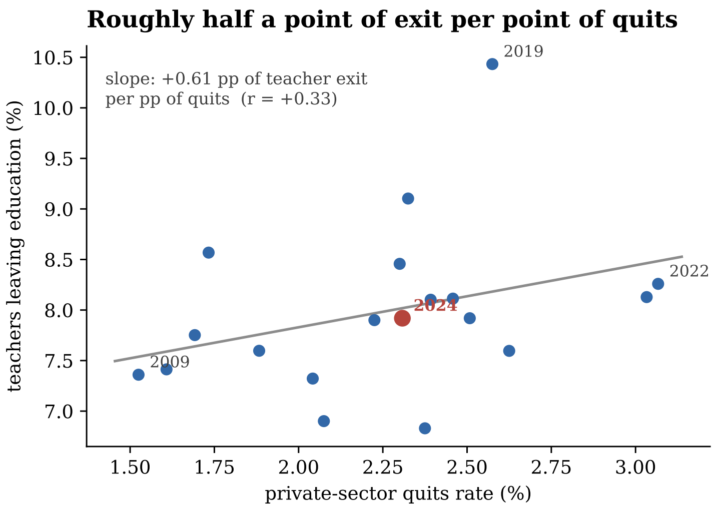

Figure 6: Teacher exit and the private-sector quits rate, 2005–2024.

Each point is a base year; OLS fit. The 2019 point carries the COVID shock; 2024, highlighted, sits where the cold market predicts.

A note on the data. Teacher series: CPS basic monthly microdata, January 2005 to December 2025 (Census Bureau; NBER mirror before 2021). The CPS re-interviews each address twelve months apart; persons are linked across interviews and validated on sex, race and age progression, following Madrian and Lefgren (2000). Attrition (Figures 2, 3, 5 and 6) is measured on public-school full-time K–12 teachers, occupations 2310, 2320 and 2330, with a bachelor's degree or higher, in base rotations that allow re-interviews. A leaver has left the classroom (dashed line) or every education-adjacent occupation (solid line) at t+12 and is not observed teaching in any later interview; this definition matches the public-school leaver rates of 7.6 to 8.0

percent in the NCES Teacher Follow-up Survey and the retrospective CPS literature (Harris and Adams, 2007; Aldeman and Yi, 2025). Stocks and earnings use the full K–12 universe including private schools, with final person weights throughout. External series: NCES Digest table 203.10 for enrollment; BLS via FRED for comparator earnings (LEU0252918500, LEU0252881600), quits (JTS1000QUR), unemployment (UNRATE) and the CPI-U deflator. No CPS was collected in October 2025 because of the federal shutdown, so 2025 stocks average eleven months and the 2024–25 links exclude October. Replication code: `src/30_borrowed_recovery_data.py` and `src/31_borrowed_recovery_figures.py`.

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